

Multipliers on Dirichlet type spaces

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Abstract. In this article we examine Dirichlet type spaces in the unit polydisc, and multipliers between these spaces. These results extend the corresponding work of G. D. Taylor in the unit disc.

In addition, we consider functions on the polydisc whose restrictions to lower dimensional polydiscs lie in the corresponding Dirichlet type spaces. We see that such functions need not be in the Dirichlet type space of the whole polydisc. Similar observations are made regarding multipliers.

1. Introduction

In [5] and [6] G. D. Taylor studied the Dirichlet type spaces, D_α , on the unit disc in $\mathbb{C}$. Additionally, he studied the multipliers between such spaces: functions multiplying one space D_α to another space D_β . An almost complete characterization of multipliers was achieved.

In this article we generalize Taylor's results to the case of Dirichlet type spaces on the unit polydisc in $\mathbb{C}^n$. Following Taylor, we identify the power series of holomorphic functions in D_α with elements in a weighted ℓ^2 space. As our power series are multidimensional, we allow the weights to be multidimensional as well. That is to say, the weight may vary from coordinate direction to coordinate direction.

An interesting phenomenon arises when we restrict a function in $D_{(\alpha_1, \dots, \alpha_n)}$ on the unit polydisc to a slice - a lower dimensional polydisc parallel to the coordinate axes. Such a restricted function lies in the appropriate weighted ℓ^2 space of

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the lower dimensional polydisc. The converse to this fact is, however, not true. In fact, a function may be in the appropriate weighted ℓ^2 space in each direction, without being in the corresponding weighted ℓ^2 space on the full dimensional polydisc. This implies that Taylor's results cannot be carried through to the higher dimensional case merely by examining restrictions to slices. These ideas are explored in Section 4.

In the one dimensional setting the Dirichlet type spaces are parameterized by $\alpha \in \mathbb{R}$. Much of Taylor's work relies on carefully partitioning $\mathbb{R}$ and examining how the spaces, and the multipliers between them, vary as the parameter α changes, and moves from one element of the partition of $\mathbb{R}$ to another.

As our weights may vary from coordinate to coordinate, our parameter space is $\mathbb{R}^n$, rather than $\mathbb{R}$. We can "partition" our parameter space in much the same way that Taylor does. That is to say, we can insist that we look at those parameter vectors where all the entries of the vector lie in one element of Taylor's partition of $\mathbb{R}$. If we do so, Taylor's results carry through. In fact, the same techniques of proof can be employed in many instances, although there are cases where alternate methods of proof need to be found.

The generalizations of Taylor's results to higher dimension can be found in Section 2, which examines the spaces D_α , and in Section 3, which examines multipliers between spaces.

For ease of notation we state and prove our results for the bidisc. Most proofs carry through to higher dimensional polydiscs with only the obvious changes necessary to account for differences in dimension. In the few instances where this is not the case, we comment on the changes needed for the proof in the higher dimensional case.

2. The Dirichlet type spaces, D_α

Our objects of study are holomorphic functions on the bidisc.

Definition 2.1. (Holomorphic Function) Let $\mathbb{U}^2$ denote the unit bidisc. We say that a function $f: \mathbb{U}^2 \rightarrow \mathbb{C}$ is *holomorphic* if it is holomorphic in each variable separately. Each holomorphic function, f , on the bidisc can be represented as a power series in $\mathbb{U}^2$:

$$f(z, w) = \sum_{(k, l) \in \mathbb{Z}_+^2} a_{k, l} z^k w^l,$$

with $(z, w) \in \mathbb{U}^2$, and $a_{k, l} \in \mathbb{C}$.

We examine holomorphic functions satisfying a growth condition on the coefficients of their Taylor series expansions.

Definition 2.2. (D_α) Let $\alpha = (\alpha_1, \alpha_2)$ be an element of $\mathbb{R}^2$. We define the space D_α as

$$D_\alpha = \left\{ f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^k w^l; \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2} < \infty \right\}.$$

Remark 2.3. As we shall see in Lemma 2.15, $D_{(0,0)}$ is the Hardy space $H^2(\mathbb{U}^2)$, and $D_{(-1,-1)}$ is the Bergman space on $\mathbb{U}^2$.

It is often useful to view D_α as a weighted ℓ^2 space, with inner product defined as follows.

Definition 2.4. ($\|f\|_\alpha$) Let $f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^k w^l$ and $g(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} b_{k,l} z^k w^l$ be two elements of D_α . We define the *inner product* of f and g as

$$(f, g)_\alpha = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} \overline{b_{k,l}} (k+1)^{\alpha_1} (l+1)^{\alpha_2}.$$

The *norm* of an element, f , in D_α is denoted by $\|f\|_\alpha$ and given as

$$\|f\|_\alpha = (f, f)_\alpha^{1/2} = \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |f_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2} \right)^{1/2}.$$

Definition 2.2 does not stipulate that the elements of D_α are holomorphic, and it is not, in fact, a priori clear that the functions in D_α are holomorphic. We have, however, the following proposition.

Proposition 2.5. Let $\alpha = (\alpha_1, \alpha_2) \in \mathbb{R}^2$. If $\{a_{k,l}\}_{(k,l) \in \mathbb{Z}_+^2}$ is a sequence in $\mathbb{C}$ satisfying

$$\sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2} < \infty,$$

then $f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^k w^l$ converges uniformly on compact subsets in $\mathbb{U}^2$, and is thus holomorphic on $\mathbb{U}^2$.

The proof of Proposition 2.5 relies on Abel's Lemma.

Lemma 2.6. (Abel's Lemma). *Let $\{a_{k,l}\}_{(k,l) \in \mathbb{Z}_+^2}$ be a sequence in $\mathbb{C}$, and let (z_0, w_0) be a point in $\mathbb{C}^2$. Assume that*

$$\sup_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l} z_0^k w_0^l| < M$$

for some $M < \infty$. Then

$$\sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^k w^l$$

converges uniformly on compact subsets in $|z_0|\mathbb{U} \times |w_0|\mathbb{U}$.

Proof of Proposition 2.5. We note first that the convergence of

$$\sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2}$$

implies that the terms $|a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2}$ are bounded.

Next, we fix $0 \leq r_1, r_2 < 1$ and show that

$$\sup_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}| r_1^k r_2^l < M < \infty.$$

Then, by Abel's Lemma, we have that $f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^k w^l$ converges uniformly on compact subsets in the bidisc $r_1\mathbb{U} \times r_2\mathbb{U}$. Since r_1 and r_2 were arbitrary, we conclude that $f(z, w)$ converges uniformly on compact subsets in $\mathbb{U}^2$.

There exists K such that for $k, l > K$ we have

$$(1) \quad r_1^k r_2^l < (k+1)^{\alpha_1/2} (l+1)^{\alpha_2/2}.$$

Thus $|a_{k,l}| r_1^k r_2^l < |a_{k,l}| (k+1)^{\alpha_1/2} (l+1)^{\alpha_2/2}$, for all $k, l > K$.

From the remark at the beginning of this proof we see that the numbers $|a_{k,l}| (k+1)^{\alpha_1/2} (l+1)^{\alpha_2/2}$ are bounded. Hence, $|a_{k,l}| r_1^k r_2^l$ is bounded for $k, l > K$.

There exists N_1 such that if $k > N_1$ and $0 \leq l \leq K$, then (1) holds. Reasoning as above, we conclude that $|a_{k,l}| r_1^k r_2^l$ is bounded for $0 \leq l \leq K$ and $k > N_1$. An analogous argument gives us the existence of N_2 such that $|a_{k,l}| r_1^k r_2^l$ is bounded for $l > N_2$ and $0 \leq k \leq K$.

Only finitely many terms have not been estimated: $|a_{k,l}| r_1^k r_2^l$, with $0 \leq k \leq N_1$ and $0 \leq l \leq N_2$. These are, of course, bounded. We conclude that there is an $M < \infty$ such that $\sup_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}| r_1^k r_2^l < M$. ■

We next define a useful linear functional on D_α : point evaluation.

Definition 2.7. ($\lambda_{(z,w)}^\alpha$) Let f be an element of D_α . We define $\lambda_{(z,w)}^\alpha(f)$ as

$$\lambda_{(z,w)}^\alpha(f) = f(z, w).$$

While it is clear that $\lambda_{(z,w)}^\alpha$ is a linear functional on D_α , we will need to use the following technical lemma to show that it is bounded.

Lemma 2.8. For all (z, w) in $\mathbb{U}^2$, the sum

$$\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2}$$

converges.

Proof. Fix (z, w) in $\mathbb{U}^2$. There exists K such that for $k, l > K$ we have

$$|z|^{2k} |w|^{2l} < (k+1)^{\alpha_1-2} (l+1)^{\alpha_2-2}.$$

Thus we have

$$(2) \quad |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} < (k+1)^{-2} (l+1)^{-2}$$

for $k, l > K$.

For $0 \leq l \leq K$, there exists N_1 such that if $k > N_1$, then (2) holds. Analogously, for $0 \leq k \leq K$, there exists N_2 so that if $l > N_2$, then (2) holds.

Let $N = \max\{K, N_1, N_2\}$. Then

$$\begin{aligned} & \sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \\ & \leq \sum_{0 \leq l, k \leq N} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} + \sum_{(k,l) \in \mathbb{Z}_+^2} (k+1)^{-2} (l+1)^{-2}. \end{aligned}$$

Since $\sum_{(k,l) \in \mathbb{Z}_+^2} (k+1)^{-2} (l+1)^{-2}$ converges, we have that

$$\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2}$$

converges. ■

Lemma 2.9. *The functional $\lambda_{(z,w)}^\alpha$ is a bounded linear functional on D_α , with norm*

$$\left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2}.$$

Proof. Let $f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^k w^l \in D_\alpha$. Then

$$\begin{aligned} |\lambda_{(z,w)}^\alpha(f)| &= |f(z, w)| \\ &\leq \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}| |z|^k |w|^l \\ &= \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}| |z|^k |w|^l (k+1)^{\alpha_1/2} (l+1)^{\alpha_2/2} (k+1)^{-\alpha_1/2} (l+1)^{-\alpha_2/2} \\ &\leq \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2} \right)^{1/2} \\ &\quad \cdot \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2} \\ &= \|f\|_\alpha \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2}. \end{aligned}$$

We see that $\|\lambda_{(z,w)}^\alpha\| \leq \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2}$. Let (z_0, w_0) be in $\mathbb{U}^2$, and define $K_{(z_0, w_0)}^\alpha: \mathbb{U}^2 \rightarrow \mathbb{C}$ by

$$K_{(z_0, w_0)}^\alpha(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} \overline{z_0}^k \overline{w_0}^l z^k w^l (k+1)^{-\alpha_1} (l+1)^{-\alpha_2}.$$

We note that $K_{(z_0, w_0)}^\alpha$ is in D_α since

$$\|K_{(z_0, w_0)}^\alpha\|_\alpha^2 = \sum_{(k,l) \in \mathbb{Z}_+^2} |z_0|^{2k} |w_0|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2}$$

converges, by Lemma 2.8.

Now,

$$\begin{aligned}
 |\lambda_{(z,w)}^\alpha(K_{(z,w)}^\alpha)| &= \sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \\
 &= \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2} \\
 &\quad \cdot \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2} \\
 &= \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2} \|K_{(z,w)}^\alpha\|_\alpha.
 \end{aligned}$$

$$\text{Hence } \|\lambda_{(z,w)}^\alpha\| = \left(\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2}.$$

■

Remark 2.10. In the above proof, $K_{(z,w)}^\alpha$ is the reproducing kernel for D_α .

In order to characterize and compare Dirichlet type spaces, we define a partial ordering on our multiindices.

Definition 2.11. ($\alpha \succ \beta$, $\alpha \succeq \beta$) Let $\alpha = (\alpha_1, \alpha_2)$ and $\beta = (\beta_1, \beta_2)$. We write $\alpha \succ \beta$ if $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, and $\alpha \succeq \beta$ if $\alpha_1 \geq \beta_1$ and $\alpha_2 \geq \beta_2$.

We make some basic observations about Dirichlet type spaces, and the relationships between them.

Proposition 2.12. If $\alpha \succ (1, 1)$, then $D_\alpha \subseteq H^\infty(\mathbb{U}^2)$: the set of bounded holomorphic functions on $\mathbb{U}^2$.

Proof. We have that

$$\sum_{(k,l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \leq \sum_{(k,l) \in \mathbb{Z}_+^2} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2}.$$

This last sum is finite if $\alpha_1, \alpha_2 > 1$. This observation, together with Lemma 2.9, finishes the proof.

■

We remark that the containment here must be strict, as D_α is a Hilbert space while $H^\infty(\mathbb{U})$ is not.

It is clear that if $\alpha \succeq \beta$, then $D_\alpha \subseteq D_\beta$. If $\alpha \succ \beta$ we have strict containment.

Proposition 2.13. *If $\beta \prec \alpha$ then $D_\alpha \subset D_\beta$.*

Proof. Let $f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} (k+1)^{(-\alpha_1-1)/2} (l+1)^{(-\alpha_2-1)/2} z^k w^l$. We see that

$$\|f\|_\beta^2 = \sum_{(k,l) \in \mathbb{Z}_+^2} (k+1)^{-\alpha_1+\beta_1-1} (l+1)^{-\alpha_2+\beta_2-1}.$$

Since $-\alpha_j + \beta_j - 1 < -1$ for $j = 1, 2$, this last sum is finite, and we conclude that f is in D_β .

On the other hand, $\|f\|_\alpha^2 = \sum_{(k,l) \in \mathbb{Z}_+^2} (k+1)^{-1} (l+1)^{-1} = \infty$. Thus f is not in D_α , and we conclude that $D_\alpha \subset D_\beta$. ■

Remark 2.14. We can, in fact, say more. If $\alpha = (\alpha_1, \alpha_2) \succeq \beta = (\beta_1, \beta_2)$, with $\alpha_2 = \beta_2$ but $\alpha_1 > \beta_1$, then $D_\alpha \subset D_\beta$. To see this, consider the function

$$f(z, w) = \sum_{k \in \mathbb{Z}_+} (k+1)^{(-\alpha_1-1)/2} z^k.$$

Just as in Proposition 2.13, we see that f is in D_β but not in D_α .

We conclude this section by noting that if $\alpha \prec (0, 0)$ or $\alpha = (0, 0)$ then the norm $\|f\|_\alpha$ is equivalent to an integral. The space $D_{(0,0)}$ is the Hardy space $H^2(\mathbb{U}^2)$, and thus has norm

$$(3) \quad \|f\|_2 = \left(\frac{1}{(2\pi)^2} \sup_{0 \leq r < 1} \int_0^{2\pi} \int_0^{2\pi} |f(re^{i\theta_1}, re^{i\theta_2})|^2 d\theta_1 d\theta_2 \right)^{1/2}.$$

(See, e.g. [4].) This is, in fact, exactly the same as $\|f\|_{(0,0)}$. If $\alpha \prec (0, 0)$ we have the following.

Lemma 2.15. *For $\alpha \prec (0, 0)$ the norm, $\|f\|_\alpha$, of D_α is equivalent to the integral*

$$\left(\frac{1}{\pi^2} \int_0^1 \int_0^1 \int_0^{2\pi} \int_0^{2\pi} |f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 \frac{r_1}{(1-r_1^2)^{1+\alpha_1}} \frac{r_2}{(1-r_2^2)^{1+\alpha_2}} d\theta_1 d\theta_2 dr_1 dr_2 \right)^{1/2}$$

Proof. We examine the above integral:

$$\begin{aligned}
& \frac{1}{\pi^2} \int_0^1 \int_0^1 \int_0^{2\pi} \int_0^{2\pi} |f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 \frac{r_1}{(1-r_1^2)^{1+\alpha_1}} \frac{r_2}{(1-r_2^2)^{1+\alpha_2}} d\theta_1 d\theta_2 dr_1 dr_2 \\
&= 4 \int_0^1 \int_0^1 \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 \frac{r_1^{2k+1}}{(1-r_1^2)^{1+\alpha_1}} \frac{r_2^{2l+1}}{(1-r_2^2)^{1+\alpha_2}} dr_1 dr_2 \\
&= 4 \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 \int_0^1 \int_0^1 \frac{r_1^{2k+1}}{(1-r_1^2)^{1+\alpha_1}} \frac{r_2^{2l+1}}{(1-r_2^2)^{1+\alpha_2}} dr_1 dr_2 \\
&= 4 \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (-\alpha_1)^{-1} \binom{k-\alpha_1}{-\alpha_1}^{-1} (-\alpha_2)^{-1} \binom{l-\alpha_2}{-\alpha_2}^{-1},
\end{aligned}$$

where the last equality follows from an integration by parts.

Hardy [2, Chapter 5] shows that $(-\alpha_1)^{-1} \binom{k-\alpha_1}{-\alpha_1}^{-1}$ is asymptotic to $(k+1)^{\alpha_1}$. Using this fact and an argument similar to that in the proof of Proposition 2.5, we conclude that

$$4 \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (-\alpha_1)^{-1} \binom{k-\alpha_1}{-\alpha_1}^{-1} (-\alpha_2)^{-1} \binom{l-\alpha_2}{-\alpha_2}^{-1}$$

is comparable to $\sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2}$. ■

Remark 2.16. We note that in the above proof we can in fact allow $\alpha_1 = 0$ and $\alpha_2 < 0$. To see this, we replace the integral

$$\frac{1}{\pi} \int_0^{2\pi} \int_0^1 |f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 \frac{r_1}{(1-r_1^2)^{1+\alpha_1}} dr_1 d\theta_1$$

with the supremum $\frac{1}{2\pi} \sup_{0 \leq r_1 < 1} \int_0^{2\pi} |f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 d\theta_1$. The proof then proceeds in essentially the same manner as above.

We conclude that for $\alpha \preceq (0, 0)$, the norm of D_α is equivalent to an integral.

3. Multipliers

A multiplier from D_α to D_β is a function which acts by pointwise multiplication as a mapping from D_α to D_β . We remark that multipliers often arise in functional analytic contexts; for example in the theory of integral and differential operators.

We provide a precise definition.

Definition 3.1. (Multiplier, $\mathcal{M}(D_\alpha, D_\beta)$) Let $\mathcal{M}(D_\alpha, D_\beta)$ be defined as

$$\mathcal{M}(D_\alpha, D_\beta) = \{\phi: \mathbb{U}^2 \rightarrow \mathbb{C}; \phi f \in D_\beta, \text{ for all } f \in D_\alpha\}.$$

We write $\mathcal{M}(D_\alpha)$ for $\mathcal{M}(D_\alpha, D_\alpha)$. An element of $\mathcal{M}(D_\alpha, D_\beta)$ (resp. $\mathcal{M}(D_\alpha)$) is called a *multiplier from D_α to D_β* (resp. a *multiplier of D_α*).

Our goal is to obtain as complete a characterization as possible of the spaces $\mathcal{M}(D_\alpha, D_\beta)$ and $\mathcal{M}(D_\alpha)$. We begin with some basic remarks about multipliers.

We note first that since 1 is in D_α for all α in $\mathbb{R}^2$, we have that $\mathcal{M}(D_\alpha, D_\beta) \subseteq D_\beta$. Next, for h in $\mathcal{M}(D_\alpha, D_\beta)$, we define $T_h: D_\alpha \rightarrow D_\beta$ by $T_h f = hf$, for f in D_α . It is clear that T_h is a linear transformation; in fact, T_h is bounded.

Lemma 3.2. *For h in $\mathcal{M}(D_\alpha, D_\beta)$, T_h is a bounded linear transformation.*

Lemma 2.9 states that point evaluation is a bounded linear functional on D_α . It is well known that a multiplier between two Banach spaces where point evaluation is bounded is a bounded linear transformation. See, e.g. [6, Chapter 1, Section 1].

We denote the norm of T_h by $\|T_h\|_{\alpha, \beta}$, suppressing the α and β , if they are clear from the context. The lemma tells us that given $h \in \mathcal{M}(D_\alpha, D_\beta)$, we obtain a bounded linear transformation. We have, in addition, pointwise estimates on elements of $\mathcal{M}(D_\alpha, D_\beta)$.

Lemma 3.3. *Let h be in $\mathcal{M}(D_\alpha, D_\beta)$. Then*

$$|h(z, w)| \leq \|T_h\|_{\alpha, \beta} \frac{\|\lambda_{(z, w)}^\beta\|}{\|\lambda_{(z, w)}^\alpha\|}.$$

This lemma is also a well-known result, following from the boundedness of point evaluation. See again [6, Chapter 1, Section 1].

As a simple consequence of this lemma, we obtain that if $\alpha \preceq \beta$ then the multipliers from D_α to D_β are bounded analytic functions.

Corollary 3.4. *If $\alpha \preceq \beta$ then $\mathcal{M}(D_\alpha, D_\beta) \subseteq H^\infty(\mathbb{U}^2)$. In particular, $\mathcal{M}(D_\alpha) \subseteq H^\infty(\mathbb{U}^2)$.*

Proof. If $\alpha \preceq \beta$ then $\|\lambda_{(z,w)}^\beta\| \leq \|\lambda_{(z,w)}^\alpha\|$. Applying Lemma 3.3 yields the result. ■

Remark 3.5. We shall see later (Theorem 3.13 and Proposition 4.11) that $M(D_\alpha, D_\beta) = \{0\}$ if $\alpha \prec \beta$.

The following interpolation result will be central in some of our later estimates.

Theorem 3.6. Let $\alpha^1 = (\alpha_1^1, \alpha_2^1)$ and $\beta^1 = (\beta_1^1, \beta_2^1)$, and $\alpha^2 = (\alpha_1^2, \alpha_2^2)$ and $\beta^2 = (\beta_1^2, \beta_2^2)$. Let

$$\begin{aligned}\alpha &= (\alpha_1, \alpha_2) = ((1-\lambda)\alpha_1^1 + \lambda\alpha_1^2, (1-\lambda)\alpha_2^1 + \lambda\alpha_2^2), \\ \beta &= (\beta_1, \beta_2) = ((1-\lambda)\beta_1^1 + \lambda\beta_1^2, (1-\lambda)\beta_2^1 + \lambda\beta_2^2).\end{aligned}$$

If $h \in \mathcal{M}(D_{\alpha^1}, D_{\beta^1}) \cap \mathcal{M}(D_{\alpha^2}, D_{\beta^2})$, then $h \in \mathcal{M}(D_\alpha, D_\beta)$ and

$$\|T_h\|_{\alpha, \beta} \leq \|T_h\|_{\alpha^1, \beta^1}^{1-\lambda} \|T_h\|_{\alpha^2, \beta^2}^\lambda.$$

The proof of Theorem 3.6 is essentially the same as Taylor's proof of the analogous result in the one dimensional case, except that we are interpolating between points in $\mathbb{R}^2$ rather than between points in $\mathbb{R}$. The proofs are, in fact, so similar that we refer the reader to Taylor's proof [6, Chapter 1, Section 3]. We point out, as Taylor did in [5], that the proof is very similar to the proof of the Riesz–Thorin Interpolation Theorem.

Further, following a suggestion from the referee, we take a moment to comment on “geometric interpolation spaces”. The idea is as follows. Let $(\mathcal{K}_0, \mathcal{L}_0) = (D_{\alpha^1}, D_{\beta^1})$, $(\mathcal{K}_1, \mathcal{L}_1) = (D_{\alpha^2}, D_{\beta^2})$, and T be a bounded linear transformation that maps $\mathcal{K}_0$ to $\mathcal{L}_0$ and $\mathcal{K}_1$ to $\mathcal{L}_1$. We would like to find a scale of Hilbert spaces $(\mathcal{K}_\lambda, \mathcal{L}_\lambda)$, $(0 < \lambda < 1)$, connecting $(\mathcal{K}_0, \mathcal{L}_0)$ and $(\mathcal{K}_1, \mathcal{L}_1)$, such that T maps $\mathcal{K}_\lambda$ to $\mathcal{L}_\lambda$ and

$$\|T\|_{\mathcal{K}_\lambda \rightarrow \mathcal{L}_\lambda} \leq \|T\|_{\mathcal{K}_0 \rightarrow \mathcal{L}_0}^{1-\lambda} \|T\|_{\mathcal{K}_1 \rightarrow \mathcal{L}_1}^\lambda.$$

The spaces $(\mathcal{K}_\lambda, \mathcal{L}_\lambda)$ are called geometric interpolation spaces. In [1, Appendix C] it is shown that if $\mathcal{K}_0$ and $\mathcal{K}_1$ are “compatible”, as are $\mathcal{L}_0$ and $\mathcal{L}_1$, then the geometric interpolation spaces exist, are unique, and a method to construct them is given. Following this method, one sees that in our case the geometric interpolation spaces are precisely the (D_α, D_β) , where

$$\alpha = (\alpha_1, \alpha_2) = ((1-\lambda)\alpha_1^1 + \lambda\alpha_1^2, (1-\lambda)\alpha_2^1 + \lambda\alpha_2^2)$$

and

$$\beta = (\beta_1, \beta_2) = ((1-\lambda)\beta_1^1 + \lambda\beta_1^2, (1-\lambda)\beta_2^1 + \lambda\beta_2^2),$$

with $0 < \lambda < 1$.

Remark 3.7. We also easily see that if $h \in \mathcal{M}(D_\alpha, D_\beta)$, then $h \in \mathcal{M}(D_\alpha, D_\mu)$ for $\mu \preceq \beta$. This follows from the fact that $D_\beta \subseteq D_\mu$ if $\mu \preceq \beta$.

Similarly, if $h \in \mathcal{M}(D_\alpha, D_\beta)$, then $h \in \mathcal{M}(D_\tau, D_\beta)$ for $\tau \succeq \alpha$.

We are now in a position to characterize some of the spaces $\mathcal{M}(D_\alpha, D_\beta)$.

Theorem 3.8. *Let $(0, 0) \succ \alpha \succeq \beta$. Then $h \in \mathcal{M}(D_\alpha, D_\beta)$ if and only if*

$$|h(z, w)| \leq C \left((1 - |z|^2)^{(\beta_1 - \alpha_1)/2} (1 - |w|^2)^{(\beta_2 - \alpha_2)/2} \right).$$

Proof. Suppose that h is in $\mathcal{M}(D_\alpha, D_\beta)$. Then

$$\begin{aligned} |h(z, w)| &\leq \|T_h\| \frac{\|\lambda_{(z, w)}^\beta\|}{\|\lambda_{(z, w)}^\alpha\|} \\ &= \|T_h\| \frac{\left(\sum_{(k, l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\beta_1} (l+1)^{-\beta_2} \right)^{1/2}}{\left(\sum_{(k, l) \in \mathbb{Z}_+^2} |z|^{2k} |w|^{2l} (k+1)^{-\alpha_1} (l+1)^{-\alpha_2} \right)^{1/2}} \\ &= \|T_h\| \left(\frac{\sum_{k \in \mathbb{Z}_+} |z|^{2k} (k+1)^{-\beta_1} \sum_{l \in \mathbb{Z}_+} |w|^{2l} (l+1)^{-\beta_2}}{\sum_{k \in \mathbb{Z}_+} |z|^{2k} (k+1)^{-\alpha_1} \sum_{l \in \mathbb{Z}_+} |w|^{2l} (l+1)^{-\alpha_2}} \right)^{1/2}. \end{aligned}$$

Terms of the form $(k+1)^{-\beta_1}$ are comparable to terms of the form $k^{-\beta_1}$, which, for k large, are comparable [2, Chapter 5] to terms of the form $\binom{k-\beta_1}{-\beta_1}$. The sum

$$\sum_{k \in \mathbb{Z}_+} \binom{k-\beta_1}{-\beta_1} |z|^{2k},$$

with $1 > \beta_1$, is comparable [2, Chapter 5] to $(1 - |z|^2)^{\beta_1-1}$. We conclude that $|h(z, w)|$ is comparable to $(1 - |z|^2)^{(\beta_1 - \alpha_1)/2} (1 - |w|^2)^{(\beta_2 - \alpha_2)/2}$.

Assume conversely that $|h(z, w)|$ is comparable to $(1 - |z|^2)^{(\beta_1 - \alpha_1)/2} (1 - |w|^2)^{(\beta_2 - \alpha_2)/2}$, and let f be a function in D_α . We recall that the D_β norm of hf is

equivalent to

$$\begin{aligned}
& \frac{1}{\pi^2} \int_0^1 \int_0^1 \int_0^{2\pi} \int_0^{2\pi} |h(r_1 e^{i\theta_1}, r_2 e^{i\theta_2}) f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 \\
& \quad \frac{r_1}{(1-r_1^2)^{1+\beta_1}} \frac{r_2}{(1-r_2^2)^{1+\beta_2}} d\theta_1 d\theta_2 dr_1 dr_2 \\
& \leq C \frac{1}{\pi^2} \int_0^1 \int_0^1 \int_0^{2\pi} \int_0^{2\pi} |f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 (1-r_1^2)^{\beta_1-\alpha_1} (1-r_2^2)^{\beta_2-\alpha_2} \\
& \quad \frac{r_1}{(1-r_1^2)^{1+\beta_1}} \frac{r_2}{(1-r_2^2)^{1+\beta_2}} d\theta_1 d\theta_2 dr_1 dr_2 \\
& = \frac{C}{\pi^2} \int_0^1 \int_0^1 \int_0^{2\pi} \int_0^{2\pi} |f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 \frac{r_1}{(1-r_1^2)^{1+\alpha_1}} \frac{r_2}{(1-r_2^2)^{1+\alpha_2}} d\theta_1 d\theta_2 dr_1 dr_2.
\end{aligned}$$

This last integral is finite, as it is equivalent to the D_α norm of f . We conclude that h is in $M(D_\alpha, D_\beta)$. ■

If $\alpha \preceq (0, 0)$ we have a complete characterization of $\mathcal{M}(D_\alpha)$.

Proposition 3.9. *If $\alpha \preceq (0, 0)$ then $\mathcal{M}(D_\alpha) = H^\infty(\mathbb{U}^2)$.*

Proof. We have seen that $\mathcal{M}(D_\alpha) \subseteq H^\infty(\mathbb{U}^2)$. It is clear from the integral representation of the norm of D_α , with $\alpha \preceq (0, 0)$, that we have $H^\infty(\mathbb{U}^2) \subseteq \mathcal{M}(D_\alpha)$. ■

We also have a complete characterization of multipliers from D_α , $\alpha \succ (1, 1)$, to D_β , $\beta \preceq \alpha$.

Theorem 3.10. *Let $\alpha \succ (1, 1)$, $\beta \preceq \alpha$. Then $\mathcal{M}(D_\alpha, D_\beta) = D_\beta$.*

Proof. We have the inclusion $\mathcal{M}(D_\alpha, D_\beta) \subseteq D_\beta$. To prove the inclusion $D_\beta \subseteq \mathcal{M}(D_\alpha, D_\beta)$, we let f be an element of D_α and g be an element of D_β :

$$\begin{aligned}
f(z, w) &= \sum_{(k, l) \in \mathbb{Z}_+^2} a_{k, l} z^k w^l, \text{ and} \\
g(z, w) &= \sum_{(k, l) \in \mathbb{Z}_+^2} b_{k, l} z^k w^l.
\end{aligned}$$

Then

$$(f \cdot g)(z, w) = \sum_{(k, l) \in \mathbb{Z}_+^2} \left(\sum_{0 \leq m \leq k} \sum_{0 \leq n \leq l} a_{m, n} b_{k-m, l-n} \right) z^k w^l,$$

and

$$\|fg\|_\beta^2 = \sum_{(k,l) \in \mathbb{Z}_+^2} (k+1)^{\beta_1} (l+1)^{\beta_2} \left| \sum_{0 \leq m \leq k} \sum_{0 \leq n \leq l} a_{m,n} b_{k-m, l-n} \right|^2.$$

For ease of notation, let $\sum'_{k,l}$ denote $\sum_{0 \leq m \leq k} \sum_{0 \leq n \leq l}$. We multiply each term, $a_{m,n} b_{k-m, l-n}$, by

$$\frac{(m+1)^{\alpha_1/2} (n+1)^{\alpha_2/2} (k-m+1)^{\beta_1/2} (l-n+1)^{\beta_2/2}}{(m+1)^{\alpha_1/2} (n+1)^{\alpha_2/2} (k-m+1)^{\beta_1/2} (l-n+1)^{\beta_2/2}},$$

and applying the Cauchy–Schwarz inequality obtain

$$\begin{aligned} (4) \quad & \|fg\|_\beta^2 \\ & \leq \sum_{(k,l) \in \mathbb{Z}_+^2} \left(\sum'_{k,l} \frac{1}{(m+1)^{\alpha_1} (n+1)^{\alpha_2} (k-m+1)^{\beta_1} (l-n+1)^{\beta_2}} \right) \\ & \quad \cdot \left(\sum'_{k,l} (m+1)^{\alpha_1} (n+1)^{\alpha_2} (k-m+1)^{\beta_1} (l-n+1)^{\beta_2} |a_{m,n}|^2 |b_{k-m, l-n}|^2 \right) \\ & \quad \cdot (k+1)^{\beta_1} (l+1)^{\beta_2}. \end{aligned}$$

Next, we notice that

$$\begin{aligned} (5) \quad & (k+1)^{\beta_1} \sum_{0 \leq m \leq k} \frac{1}{(m+1)^{\alpha_1} (k-m+1)^{\beta_1}} \\ & = \frac{(k+1)^{\beta_1}}{(k+2)^{\alpha_1}} \sum_{0 \leq m \leq k} \left(\frac{1}{m+1} + \frac{1}{k-m+1} \right)^{\alpha_1} \frac{1}{(k-m+1)^{\beta_1-\alpha_1}} \\ & \leq \sum_{0 \leq m \leq k} \left(\frac{1}{m+1} + \frac{1}{k-m+1} \right)^{\alpha_1} \\ & \leq \left[\left(\sum_{0 \leq m \leq k} \frac{1}{(m+1)^{\alpha_1}} \right)^{1/\alpha_1} + \left(\sum_{0 \leq m \leq k} \frac{1}{(k-m+1)^{\alpha_1}} \right)^{1/\alpha_1} \right]^{\alpha_1} \\ & \leq C_{\alpha_1}, \end{aligned}$$

with C_{α_1} a constant depending only on α_1 . The second last inequality follows from an application of Minkowski's inequality. That the sums

$$\sum_{0 \leq m \leq k} \frac{1}{(m+1)^{\alpha_1}} \quad \text{and} \quad \sum_{0 \leq m \leq k} \frac{1}{(k-m+1)^{\alpha_1}}$$

are bounded follows from the fact that $\alpha_1 > 1$.

Similarly, we obtain

$$(6) \quad (l+1)^{\beta_2} \sum_{0 \leq n \leq l} \frac{1}{(n+1)^{\alpha_1} (l-n+1)^{\beta_2}} \leq C_{\alpha_2},$$

with C_{α_2} a constant depending only on α_2 .

Since m and n are varying independently in the sum

$$(k+1)^{\beta_1} (l+1)^{\beta_2} \sum'_{k,l} \frac{1}{(m+1)^{\alpha_1} (n+1)^{\alpha_2} (k-m+1)^{\beta_1} (l-n+1)^{\beta_2}},$$

we use Inequalities 5 and 6 to estimate this product, and conclude that it is less than $C_{\alpha_1} C_{\alpha_2}$.

Returning to Inequality 4, we see that

$$\begin{aligned} \|fg\|_{\beta}^2 &\leq C_{\alpha_1} C_{\alpha_2} \sum_{(k,l) \in \mathbb{Z}_+^2} \sum'_{k,l} (m+1)^{\alpha_1} (n+1)^{\alpha_2} (k-m+1)^{\beta_1} (l-n+1)^{\beta_2} \\ &\quad \cdot |a_{m,n}|^2 |b_{k-m,l-n}|^2 \\ &= C_{\alpha_1} C_{\alpha_2} \sum_{(k,l) \in \mathbb{Z}_+^2} \sum_{(m,n) \in \mathbb{Z}_+^2} (m+1)^{\alpha_1} (n+1)^{\alpha_2} (k+1)^{\beta_1} (l+1)^{\beta_2} \\ &\quad \cdot |a_{m,n}|^2 |b_{k,l}|^2 \\ &= C_{\alpha_1} C_{\alpha_2} \|f\|_{\alpha}^2 \|g\|_{\beta}^2 < \infty. \end{aligned}$$

Since $\|fg\|_{\beta}$ is finite, we have that g is an element of $\mathcal{M}(D_{\alpha}, D_{\beta})$. ■

Corollary 3.11. *If $\alpha \succeq \beta \succ (1, 1)$, with $\alpha_1 > \beta_1$ or $\alpha_2 > \beta_2$, then $\mathcal{M}(D_{\alpha}) \subset \mathcal{M}(D_{\beta})$.*

Proof. This follows immediately from Theorem 3.10, Proposition 2.13, and Remark 2.14. ■

We have an analogous result, without proper inclusion, for any $\alpha \succ \beta$.

Theorem 3.12. *If $\alpha \succ \beta$ then $M(D_{\alpha}) \subseteq M(D_{\beta})$.*

Proof. If f is in $M(D_\alpha)$ then, by Corollary 3.4, f is in $H^\infty(\mathbb{U}^2)$. By Proposition 3.9 we have that f is in $M(D_\gamma)$ for all $\gamma \prec (0, 0)$. There exists a $\lambda \in (0, 1)$ such that $\beta - \lambda\alpha \prec (0, 0)$. Let $\gamma = (\gamma_1, \gamma_2)$ be defined as

$$\gamma = \frac{\beta - \lambda\alpha}{1 - \lambda}.$$

Since $\gamma \prec (0, 0)$ we have that f is in $M(D_\gamma)$. Applying Theorem 3.6 with α, γ and λ , we see that f is in D_β . ■

Thus far we have examined $\mathcal{M}(D_\alpha, D_\beta)$ with $\alpha \succeq \beta$. The next theorem indicates why we have chosen to examine only these indices.

Theorem 3.13. *If $\beta \succ \alpha$, then $\mathcal{M}(D_\alpha, D_\beta) = \{0\}$.*

Proof. We begin by noting that

$$\|\lambda_{(z, w)}^{(\gamma_1, \gamma_2)}\| = \|\lambda_z^{\gamma_1}\| \|\lambda_w^{\gamma_2}\|.$$

The number on the left-hand side is the norm of the point evaluation at $(z, w) \in \mathbb{U}^2$ on the space $D_{(\gamma_1, \gamma_2)}$. The two numbers on the right-hand side are the norm of the point evaluation at $z \in \mathbb{U}$ (resp. $w \in \mathbb{U}$) on the space D_{γ_1} (resp. D_{γ_2}).

Thus, by Lemma 3.3, we have that if $h \in \mathcal{M}(D_\alpha, D_\beta)$ then

$$|h(z, w)| \leq \|T_h\|_{\alpha, \beta} \frac{\|\lambda_{(z, w)}^\beta\|}{\|\lambda_{(z, w)}^\alpha\|} = \|T_h\|_{\alpha, \beta} \frac{\|\lambda_z^{\beta_1}\| \|\lambda_w^{\beta_2}\|}{\|\lambda_z^{\alpha_1}\| \|\lambda_w^{\alpha_2}\|}.$$

In the case of Dirichlet type spaces on the unit disc, Taylor [5, Theorem 4] shows that $a < b$ implies that $\mathcal{M}(D_a, D_b) = \{0\}$. An application of Taylor's arguments shows that $\|\lambda_z^{\beta_1}\|/\|\lambda_z^{\alpha_1}\| \rightarrow 0$ as $|z| \rightarrow 1$, and that $\|\lambda_w^{\beta_2}\|/\|\lambda_w^{\alpha_2}\| \rightarrow 0$ as $|w| \rightarrow 1$. We conclude that $|h(z, w)| \rightarrow 0$ as $(z, w) \rightarrow \partial\mathbb{U}^2$. By the maximum principle we conclude that $h \equiv 0$. ■

Remark 3.14. It is of interest to note that, unlike most of the theorems we prove, Theorem 3.13 does not rely on our “partition” of $\mathbb{R}^2$. That is to say, there is no requirement that α_1 and α_2 lie in the same element of Taylor's partition of $\mathbb{R}$. They are allowed to vary independently of one another.

We note as well that a more general result than Theorem 3.13 can be proved (Proposition 4.11), once we have examined slices of functions in Section 4.

4. Slices

Given a function, f , on the bidisc, we can examine the functions obtained by restricting f to the one complex dimensional slices of the bidisc parallel to the coordinate axes. Specifically, we define f_z and f_w as follows.

Definition 4.1. (f_z, f_w) Let $f(z, w)$ be a function from $\mathbb{U}^2$ to $\mathbb{C}$, and fix $w \in \mathbb{U}$. The function $f_w(z): \mathbb{U} \rightarrow \mathbb{C}$ is defined as

$$f_w(z) = f(z, w).$$

Similarly, for fixed $z \in \mathbb{U}$ we define $f_z(w): \mathbb{U} \rightarrow \mathbb{C}$ by

$$f_z(w) = f(z, w).$$

It is natural to ask whether $f \in D_{(\alpha_1, \alpha_2)}$ implies that $f_z \in D_{\alpha_2}$ and $f_w \in D_{\alpha_1}$, for all $z, w \in \mathbb{U}$. We shall see that the answer is yes, but that the converse is not necessarily true.

Theorem 4.2. *Let f be an element of $D_{(\alpha_1, \alpha_2)}$. Then $f_{w_0} \in D_{\alpha_1}$ for each $w_0 \in \mathbb{U}$, and $f_{z_0} \in D_{\alpha_2}$ for each $z_0 \in \mathbb{U}$.*

Proof. We prove that f_{w_0} is in D_{α_1} . The proof that f_{z_0} is in D_{α_2} is the same, with the obvious modifications.

Fix $w_0 \in \mathbb{U}$. Let $f(z, w) = \sum_{(k, l) \in \mathbb{Z}_+^2} a_{k, l} z^k w^l$ be an element of D_{α} . We know that

$$\|f\|_{\alpha}^2 = \sum_{(k, l) \in \mathbb{Z}_+^2} |a_{k, l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2}$$

is finite. The function $f_{w_0}(z)$ is a holomorphic function with power series expansion

$$f_{w_0}(z) = \sum_{k \in \mathbb{Z}_+} \left(\sum_{l \in \mathbb{Z}_+} a_{k, l} w_0^l \right) z^k.$$

For each $k \in \mathbb{Z}_+$ the sum $\sum_{l \in \mathbb{Z}_+} a_{k, l} w_0^l$ converges absolutely.

To show that f_{w_0} is an element of D_{α_1} , we must show that

$$\|f_{w_0}\|_{\alpha_1}^2 = \sum_{k \in \mathbb{Z}_+} \left| \sum_{l \in \mathbb{Z}_+} a_{k, l} w_0^l \right|^2 (k+1)^{\alpha_1} < \infty.$$

We have the following.

$$\begin{aligned}
& \sum_{k \in \mathbb{Z}_+} \left| \sum_{l \in \mathbb{Z}_+} a_{k,l} w_0^l \right|^2 (k+1)^{\alpha_1} \\
& \leq \sum_{k \in \mathbb{Z}_+} \left(\sum_{l \in \mathbb{Z}_+} |a_{k,l}| |w_0|^l \right)^2 (k+1)^{\alpha_1} \\
& \leq (1 - |w_0|)^{-1} \sum_{k \in \mathbb{Z}_+} \left(\sum_{l \in \mathbb{Z}_+} |a_{k,l}|^2 |w_0|^l \right) (k+1)^{\alpha_1} \\
& \leq (1 - |w_0|)^{-1} \left(C_{w_0} + \sum_{k \in \mathbb{Z}_+} \sum_{l \in \mathbb{Z}_+} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2} \right) \\
& = (1 - |w_0|)^{-1} (C_{w_0} + \|f\|_\alpha^2) < \infty.
\end{aligned}$$

The second inequality is an application of Jensen's Inequality, and the fact that $\sum_{l \in \mathbb{Z}_+} |w_0|^l = (1 - |w_0|)^{-1}$. The last inequality follows from reasoning similar to that employed in the proof of Lemma 2.8. The constant C_{w_0} is a finite number depending on w_0 . ■

As mentioned above, the converse to Theorem 4.2 is not true.

Theorem 4.3. *Fix $\alpha = (\alpha_1, \alpha_2)$. Then there exists a function $f: \mathbb{U}^2 \rightarrow \mathbb{C}$ such that $f_w \in D_{\alpha_1}$ for all w in $\mathbb{U}$, $f_z \in D_{\alpha_2}$ for all z in $\mathbb{U}$, but f is not in D_α .*

Proof. Let $f: \mathbb{U}^2 \rightarrow \mathbb{C}$ be defined as $f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^l w^k$, where

$$a_{k,l} = \sqrt{\frac{(k+1)^{1-\alpha_1} (l+1)^{1-\alpha_2}}{(k+1)^3 + (l+1)^3}}.$$

We fix w_0 in $\mathbb{U}$. Then

$$\begin{aligned}
\|f_{w_0}\|_{\alpha_1}^2 &= \sum_{k \in \mathbb{Z}_+} \left| \sum_{l \in \mathbb{Z}_+} \left(\frac{(k+1)^{1-\alpha_1} (l+1)^{1-\alpha_2}}{(k+1)^3 + (l+1)^3} \right)^{1/2} w_0^l \right|^2 (k+1)^{\alpha_1} \\
&\leq (1 - |w_0|)^{-1} \sum_{(k,l) \in \mathbb{Z}_+^2} \frac{(k+1)(l+1)}{(k+1)^3 + (l+1)^3} (l+1)^{-\alpha_2} |w_0|^l
\end{aligned}$$

by an application of Jensen's Inequality.

Note that there is an L such that for each $k \in \mathbb{Z}_+$, and for $l \geq L$, we have that

$$\begin{aligned} |w_0|^l &< (l+1)^{\alpha_2-3} + (k+1)^{-3}(l+1)^{\alpha_2} \\ &= (k+1)^{-2}(l+1)^{\alpha_2-2} \left(\frac{(k+1)^3 + (l+1)^3}{(k+1)(l+1)} \right). \end{aligned}$$

Thus, for each $k \in \mathbb{Z}_+$, and for $l \geq L$,

$$\frac{(k+1)(l+1)}{(k+1)^3 + (l+1)^3} (l+1)^{-\alpha_2} |w_0|^l < (k+1)^{-2} (l+1)^{-2}.$$

We conclude that

$$\begin{aligned} \|f_{w_0}\|_{\alpha_1}^2 &\leq (1 - |w_0|)^{-1} \cdot \left(\sum_{\substack{k \in \mathbb{Z}_+ \\ l \geq L}} (k+1)^{-2} (l+1)^{-2} + \sum_{\substack{0 \leq l \leq L \\ k \in \mathbb{Z}_+}} \frac{(k+1)(l+1)}{(k+1)^3 + (l+1)^3} (l+1)^{-\alpha_2} |w_0|^l \right). \end{aligned}$$

The first summand is clearly finite. The second summand is finite, since for each $l \in \{0, \dots, L\}$ the series

$$\sum_{k \in \mathbb{Z}_+} \frac{(k+1)(l+1)}{(k+1)^3 + (l+1)^3} (l+1)^{-\alpha_2} |w_0|^l$$

is dominated by a constant multiple of $\sum_{k \in \mathbb{Z}_+} \frac{1}{(k+1)^2}$. Hence $\|f_{w_0}\|_{\alpha_1}$ is finite, and f_{w_0} is in D_{α_1} .

We have shown that f_w is in D_{α_1} , for each $w \in \mathbb{U}$. By the symmetry of the Taylor coefficients of f we see that f_z is in D_{α_2} , for each $z \in \mathbb{U}$.

Since f_z is in D_{α_2} for each $z \in \mathbb{U}$, we see that f_z is holomorphic for each $z \in \mathbb{U}$. Similarly, f_w is holomorphic for each $w \in \mathbb{U}$. We conclude that f is holomorphic on $\mathbb{U}^2$.

The D_α norm of f is, however, infinite:

$$\|f\|_\alpha^2 = \sum_{(k,l) \in \mathbb{Z}_+^2} \frac{(k+1)(l+1)}{(k+1)^3 + (l+1)^3} > \sum_{k \in \mathbb{Z}_+} \frac{1}{2(k+1)} = \infty. \quad \blacksquare$$

Remark 4.4. We remark that analogous examples can be found in higher dimensions. Let $f: \mathbb{U}^n \rightarrow \mathbb{C}$ be defined as

$$f(z_1, \dots, z_n) = \sum_{j_1 \in \mathbb{Z}_+} \cdots \sum_{j_n \in \mathbb{Z}_+} a_{j_1, \dots, j_n} z_1^{j_1} \cdots z_n^{j_n},$$

with

$$a_{j_1, \dots, j_n} = \left(\frac{j_1^{1-\alpha_1} \cdots j_n^{1-\alpha_n}}{j_1^{2n-1} + \cdots + j_n^{2n-1}} \right)^{1/2}.$$

Then f is not in $D_{(\alpha_1, \dots, \alpha_n)}$. However, if we fix k coordinates, $1 \leq k \leq n-1$, then f is in the corresponding $n-k$ dimensional Dirichlet space. For example, fix $z_1 = b_1, \dots, z_k = b_k$, with $k \in \{1, \dots, n-1\}$. Then

$$f(b_1, \dots, b_k, z_{k+1}, \dots, z_n): \mathbb{U}^{n-k} \rightarrow \mathbb{C}$$

is in $D_{(\alpha_{k+1}, \dots, \alpha_n)}$.

Theorem 4.2 tells us that f_w is in D_{α_1} for each $w \in \mathbb{U}$. It does not, however, indicate whether the norms $\|f_w\|_{\alpha_1}$ are uniformly bounded in w .

An example illustrates that in general there need be no such bound. Consider the function $f(z, w) = \sum_{(k,l) \in \mathbb{Z}_+^2} a_{k,l} z^k w^l$, where $a_{k,l} = 1$ for all $(k, l) \in \mathbb{Z}_+^2$. Let $\alpha_1, \alpha_2 < -1$. The function f is in $D_{(\alpha_1, \alpha_2)}$, with

$$\|f\|_{(\alpha_1, \alpha_2)}^2 = \sum_{(k,l) \in \mathbb{Z}^2} (k+1)^{\alpha_1} (l+1)^{\alpha_2}.$$

On the other hand, the D_{α_1} norm of f_w is

$$\|f_w\|_{\alpha_1}^2 = \sum_{k \in \mathbb{Z}_+} \left| \sum_{l \in \mathbb{Z}_+} w^l \right|^2 (k+1)^{\alpha_1} = \left| \frac{1}{1-w} \right|^2 \sum_{k \in \mathbb{Z}_+} (k+1)^{\alpha_1} = \left| \frac{1}{1-w} \right|^2 C.$$

Clearly $\|f_w\|_{\alpha_1} \rightarrow \infty$ as $w \rightarrow 1$.

This example motivates the following question regarding a partial converse to Theorem 4.2. Let f be a function, $f: \mathbb{U}^2 \rightarrow \mathbb{C}$. Assume that $\|f_w\|_{\alpha_1} < M$ for each $w \in \mathbb{U}$. When does this imply that $f \in D_{(\alpha_1, \alpha_2)}$? For which α_2 ?

We present several results in this direction. The first result indicates that, in general, uniform boundedness in w of $\|f_w\|_{\alpha_1}$ is not enough to guarantee that f is in $D_{(\alpha_1, \alpha_2)}$. The second result, however, indicates that if we restrict our attention to a suitable set of (α_1, α_2) , then uniform boundedness in w of $\|f_w\|_{\alpha_1}$ does guarantee that f is in $D_{(\alpha_1, \alpha_2)}$. Finally, we illustrate a simple situation where a converse to Theorem 4.2 holds.

Theorem 4.5. *Let $\alpha_2 > 1$. For any $\alpha_1 \in \mathbb{R}$, there is a function $f: \mathbb{U}^2 \rightarrow \mathbb{C}$ such that $\|f_w\|_{\alpha_1} < M < \infty$ for all $w \in \mathbb{U}$, but f is not in $D_{(\alpha_1, \alpha_2)}$.*

Proof. Since $\alpha_2 > 1$ we know that $D_{\alpha_2} \subseteq H^\infty(\mathbb{U})$. Since $H^\infty(\mathbb{U})$ is not a Hilbert space, while D_{α_2} is, the containment is strict. Let g be a function in $H^\infty(\mathbb{U}) \setminus D_{\alpha_2}$, and assume that $\|g\|_{H^\infty} = 1$.

Define $g_k = (k+1)^{(-\alpha_1-2)/2}g$, so that $\|g_k\|_\infty = (k+1)^{(-\alpha_1-2)/2}$, and expand g_k as $g_k(w) = \sum_{l \in \mathbb{Z}_+} a_{k,l} w^l$. Clearly g_k is not in D_{α_2} .

Define $f(z, w)$ as $f(z, w) = \sum_{k,l \in \mathbb{Z}_+^2} a_{k,l} z^k w^l$.

Then

$$\begin{aligned} \|f\|_{(\alpha_1, \alpha_2)}^2 &= \sum_{k,l \in \mathbb{Z}_+^2} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2} = \sum_{k \in \mathbb{Z}_+} \left(\sum_{l \in \mathbb{Z}_+} |a_{k,l}|^2 (l+1)^{\alpha_2} \right) (k+1)^{\alpha_1} \\ &= \sum_{k \in \mathbb{Z}_+} \|g_k\|_{\alpha_2}^2 (k+1)^{\alpha_1} = \infty. \end{aligned}$$

On the other hand,

$$\|f_w\|_{\alpha_1}^2 = \sum_{k \in \mathbb{Z}_+} \left| \sum_{l \in \mathbb{Z}_+} a_{k,l} w^l \right|^2 (k+1)^{\alpha_1} \leq \sum_{k \in \mathbb{Z}_+} (k+1)^{-2} = C,$$

where C is a finite constant independent of w . ■

While we cannot guarantee that uniform boundedness in w of $\|f_w\|_{\alpha_1}$ implies inclusion in $D_{(\alpha_1, \alpha_2)}$, we now see that there are situations where this is the case.

Theorem 4.6. *Let $\alpha_1 \leq 0$. Suppose that $f: \mathbb{U}^2 \rightarrow \mathbb{C}$ is such that $\|f_w\|_{\alpha_1} < M < \infty$ for all $w \in \mathbb{U}$. Then, for any $\alpha_2 \leq 0$, f is in $D_{(\alpha_1, \alpha_2)}$.*

Proof. Since f_w is in D_{α_1} and $\alpha_1 \leq 0$, by Lemma 2.15 we have that $\|f_w\|_{\alpha_1}$ is comparable to

$$I_{\alpha_1}(f_w) = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 |f_w(re^{i\theta})|^2 (1-r)^{-1-\alpha_1} r dr d\theta$$

if $\alpha_1 < 0$, and comparable to $I_{\alpha_1}(f_w) = \sup_{0 \leq r < 1} \int_0^{2\pi} |f_w(re^{i\theta})|^2 d\theta$ if $\alpha_1 = 0$.

We are assuming that $\|f_w\|_{\alpha_1} < M < \infty$ for all $w \in \mathbb{U}$, so in fact $I_{\alpha_1}(f_w) < M$ for all $w \in \mathbb{U}$.

For ease of notation we assume that α_1, α_2 are both less than 0. In this case $\|f\|_{(\alpha_1, \alpha_2)}$ is equivalent to

$$\begin{aligned} & \frac{1}{\pi} \int_0^{2\pi} \int_0^1 \left(\frac{1}{\pi} \int_0^{2\pi} \int_0^1 |f(r_1 e^{i\theta_1}, r_2 e^{i\theta_2})|^2 \frac{r_1}{(1-r_1)^{1+\alpha_1}} dr_1 d\theta_1 \right) \frac{r_2}{(1-r_2)^{1+\alpha_2}} dr_2 d\theta_2 \\ & < \frac{1}{\pi} \int_0^{2\pi} \int_0^1 M \frac{r_2}{(1-r_2)^{1+\alpha_2}} dr_2 d\theta_2 < \infty. \end{aligned}$$

We conclude that f is in $D_{(\alpha_1, \alpha_2)}$.

(We remark that if α_1 or α_2 are equal to 0, then the obvious adjustments must be made, but the proof goes through in exactly the same fashion.) ■

While it is in general not the case that the converse to Theorem 4.2 is true, in the following simple situation it does hold.

Proposition 4.7. *Let $f_j: \mathbb{U} \rightarrow \mathbb{C}$ be an element of D_{α_j} , for $j = 1, 2$. Then $f(z, w) = f_1(z)f_2(w)$ is in $D_{(\alpha_1, \alpha_2)}$.*

Proof. The Taylor coefficients of f are $a_{k,l} = b_k^1 b_l^2$, where b_k^j denotes the k^{th} Taylor coefficient of f_j . We see that

$$\begin{aligned} \|f\|_{\alpha}^2 &= \sum_{(k,l) \in \mathbb{Z}_+^2} |a_{k,l}|^2 (k+1)^{\alpha_1} (l+1)^{\alpha_2} = \sum_{k \in \mathbb{Z}_+} |b_k^1|^2 (k+1)^{\alpha_1} \sum_{l \in \mathbb{Z}_+} |b_l^2|^2 (l+1)^{\alpha_2} \\ &= \|f_1\|_{\alpha_1}^2 \|f_2\|_{\alpha_2}^2 < \infty, \end{aligned}$$

and conclude that f is in $D_{(\alpha_1, \alpha_2)}$. ■

Having examined the restriction of functions in D_{α} to lower dimensional discs parallel to the coordinate axes, we do the same with multipliers. Just as the restriction, f_w , of a function was in the corresponding lower dimensional Dirichlet type space, so the restriction of a multiplier is a multiplier of the lower dimensional space.

Theorem 4.8. *Let h be an element of $\mathcal{M}(D_{(\alpha_1, \alpha_2)}, D_{(\beta_1, \beta_2)})$. Then $h_{w_0} \in \mathcal{M}(D_{\alpha_1}, D_{\beta_1})$ for each $w_0 \in \mathbb{U}$, and $h_{z_0} \in \mathcal{M}(D_{\alpha_2}, D_{\beta_2})$ for each $z_0 \in \mathbb{U}$.*

Proof. We prove that h_{w_0} is in $\mathcal{M}(D_{\alpha_1}, D_{\beta_1})$. The statement regarding h_{z_0} is proved in the same fashion.

Let h be an element of $\mathcal{M}(D_{(\alpha_1, \alpha_2)}, D_{(\beta_1, \beta_2)})$, and fix $w_0 \in \mathbb{U}$. Let $f(z): \mathbb{U} \rightarrow \mathbb{C}$ be an element of D_{α_1} , and define $\tilde{f}(z, w): \mathbb{U}^2 \rightarrow \mathbb{C}$ by $\tilde{f}(z, w) = f(z)$. Notice that $\tilde{f}$ is an element of $D_{(\alpha_1, \alpha_2)}$; in fact, $\tilde{f}$ is an element of $D_{(\alpha_1, \beta)}$, for any $\beta \in \mathbb{R}$.

Since h is in $\mathcal{M}(D_{(\alpha_1, \alpha_2)}, D_{(\beta_1, \beta_2)})$, we have that $h\tilde{f}$ is in $D_{(\beta_1, \beta_2)}$. We note that $(h\tilde{f})_{w_0} = h_{w_0}\tilde{f}_{w_0} = h_{w_0}f$, and by Theorem 4.2 we have that $(h\tilde{f})_{w_0}$ is in D_{β_1} . We conclude that h_{w_0} is an element of $\mathcal{M}(D_{\alpha_1}, D_{\beta_1})$. ■

Just as the converse to Theorem 4.2 was not in general true, we shall see that the converse to Theorem 4.8 is also not true in general. In fact, the counterexample to the converse of Theorem 4.2 (Theorem 4.3) is the key ingredient in showing that this is the case.

Proposition 4.9. *Let $\gamma \succ (1, 1)$, and let $\alpha \preceq \gamma$. Then there is a function, f , such that $f_{w_0} \in \mathcal{M}(D_{\gamma_1}, D_{\alpha_1})$, and $f_{z_0} \in \mathcal{M}(D_{\gamma_2}, D_{\alpha_2})$, for each $z_0, w_0 \in \mathbb{U}$, but f is not in $\mathcal{M}(D_\gamma, D_\alpha)$.*

Proof. By Theorem 3.10 we have that $\mathcal{M}(D_\gamma, D_\alpha) = D_\alpha$. Let f be as in the proof of Theorem 4.3, so that $f_{w_0} \in D_{\alpha_1}$, $f_{z_0} \in D_{\alpha_2}$, for each $z_0, w_0 \in \mathbb{U}$, but f is not in D_α .

We know that $f_{w_0} \in \mathcal{M}(D_{\gamma_1}, D_{\alpha_1})$ and $f_{z_0} \in \mathcal{M}(D_{\gamma_2}, D_{\alpha_2})$, by Taylor's [5, Theorem 7] one variable version of Theorem 3.10. The function f , however, is not in $\mathcal{M}(D_\gamma, D_\alpha)$, as it is not an element of D_α . ■

We present one final example, showing further evidence that a converse to Theorem 4.8 does not in general hold.

Example 4.10. Let $f(z, w) = \frac{1}{z+w-2}$. Clearly f is not in $H^\infty(\mathbb{U}^2)$, and thus f is not in $\mathcal{M}(D_\alpha)$ for any $\alpha \in \mathbb{R}^2$. The slice functions, f_{w_0} and f_{z_0} , however, are in $H^\infty(\mathbb{U})$ for each $z_0, w_0 \in \mathbb{U}$. Thus, if $\alpha \preceq (0, 0)$, then $f_{z_0} \in \mathcal{M}(D_{\alpha_2})$ and $f_{w_0} \in \mathcal{M}(D_{\alpha_1})$.

We remark as well that f is not in D_α for any $\alpha \succeq (0, 0)$. To see this we note that f has the power series expansion

$$f(z, w) = - \sum_{k \in \mathbb{Z}_+} \frac{1}{2^{k+1}} \sum_{l=0}^k \binom{k}{l} z^l w^{k-l}.$$

Thus

$$\begin{aligned}
\|f\|_\alpha^2 &= \sum_{k \in \mathbb{Z}_+} \frac{1}{2^{2(k+1)}} \sum_{l=0}^k \binom{k}{l}^2 (l+1)^{\alpha_1} (k-l+1)^{\alpha_2} \\
&\geq \sum_{k \in \mathbb{Z}_+} \frac{k+1}{2^{2(k+1)}} \left(\sum_{l=0}^k \binom{k}{l}^2 \frac{1}{k+1} \right) \\
&\geq \sum_{k \in \mathbb{Z}_+} \frac{k+1}{2^{2(k+1)}} \left(\sum_{l=0}^k \binom{k}{l} \right)^2 \frac{1}{(k+1)^2} = \sum_{k \in \mathbb{Z}_+} \frac{1}{4(k+1)} = \infty.
\end{aligned}$$

We claim, however, that f_{w_0} is in D_{α_1} , for any $w_0 \in \mathbb{U}$, and any $\alpha_1 \in \mathbb{R}$. To see this we first note that we can expand $f(z, w)$ as

$$f_w(z) = - \sum_{k \in \mathbb{Z}_+} \left(\frac{1}{2-w} \right)^{k+1} z^k.$$

We notice that $1/(|2-w_0|) < 1$. Thus there is an N such that if $n \geq N$ then

$$\left(\frac{1}{|2-w_0|} \right)^{2(n+1)} < (n+1)^{-\alpha_1-2}.$$

Then

$$\begin{aligned}
\|f_{w_0}\|_{\alpha_1}^2 &= \sum_{k \in \mathbb{Z}_+} \left(\frac{1}{|2-w_0|} \right)^{2(k+1)} (k+1)^{\alpha_1} \\
&\leq \sum_{k=0}^N \left(\frac{1}{|2-w_0|} \right)^{2(k+1)} (k+1)^{\alpha_1} + \sum_{k=N+1}^{\infty} (n+1)^{-2} < \infty,
\end{aligned}$$

and thus f_{w_0} is in D_{α_1} .

Analogous calculations show that f_{z_0} is in D_{α_2} , for any $\alpha_2 \in \mathbb{R}$, and any $z_0 \in \mathbb{U}$.

Fix $\gamma \succ (1, 1)$, and $(0, 0) \preceq (\alpha_1, \alpha_2) \preceq (\gamma_1, \gamma_2)$. We saw above that f is not in D_α , and is thus not in $\mathcal{M}(D_\gamma, D_\alpha)$. We have also seen that f_{z_0} is in D_{α_2} , and is thus in $\mathcal{M}(D_{\gamma_2}, D_{\alpha_2})$, and that f_{w_0} is in D_{α_1} , and is thus in $\mathcal{M}(D_{\gamma_1}, D_{\alpha_1})$.

We mention, finally, a simple corollary of our work with slices. We easily generalize Theorem 3.13.

Proposition 4.11. *Let β and α be elements of $\mathbb{R}^2$. If $\beta_1 > \alpha_1$ then $\mathcal{M}(D_\alpha, D_\beta) = \{0\}$, regardless of the relationship between α_2 and β_2 .*

The same is true if $\beta_2 > \alpha_2$, regardless of the relationship between α_1 and β_1 .

Proof. We recall that Taylor [5, Theorem 4] proved that if $a < b$ then $\mathcal{M}(D_a, D_b) = \{0\}$.

Assume that $\alpha_1 < \beta_1$. We know from Theorem 4.8 that if f is in $\mathcal{M}(D_\alpha, D_\beta)$ then $f_w(z)$ is in $\mathcal{M}(D_{\alpha_1}, D_{\beta_1})$ for each $w \in \mathbb{U}$. This implies that f_w is identically zero for each $w \in \mathbb{U}$, and thus that $f \equiv 0$. ■

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